Executive Summary

This report presents a sample of data-driven visualisations developed using SQL and Power BI to analyse student academic performance. These tools help teachers, counselors, and school administrators move beyond qualitative judgments and make informed decisions based on actual data. The seven visualisations shown here are just examples of what can be done. Many more advanced analyses and different scenarios can be defined to meet the specific needs of the school.

The first three charts focus on performance tracking: comparing mock and final exam averages, monitoring individual student progress across subjects, and comparing a student's grades with the class average using a Decomposition Tree. The next two charts enable multi-perspective tracking, allowing users to monitor a student's situation across several charts simultaneously and compare their performance with peers at a glance. The final two charts are particularly valuable: one compares academic performance between male and female students to identify patterns and encourage healthy competition, and the other integrates discipline records with academic grades to support students who need behavioural improvement. This is important because disruptive behaviour not only affects the student's own learning but also disturbs the classroom environment and negatively impacts other students' education.

📊 1. Average Grades – Mock vs Final Exams (All Students)

This chart compares the average grades of all students in each subject between the mock exams and the final exams.
It helps us see if students have improved or declined in each subject over time.
Also, it gives us a chance to check how reliable the mock exams are in predicting final results.

Goal: Understand overall performance trends and evaluate exam consistency.

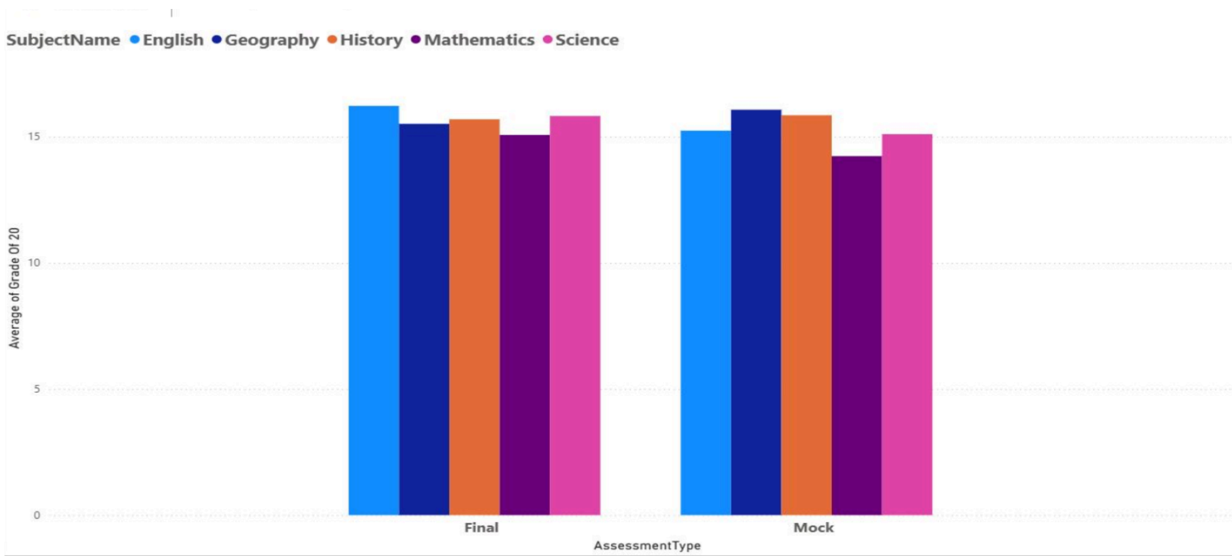
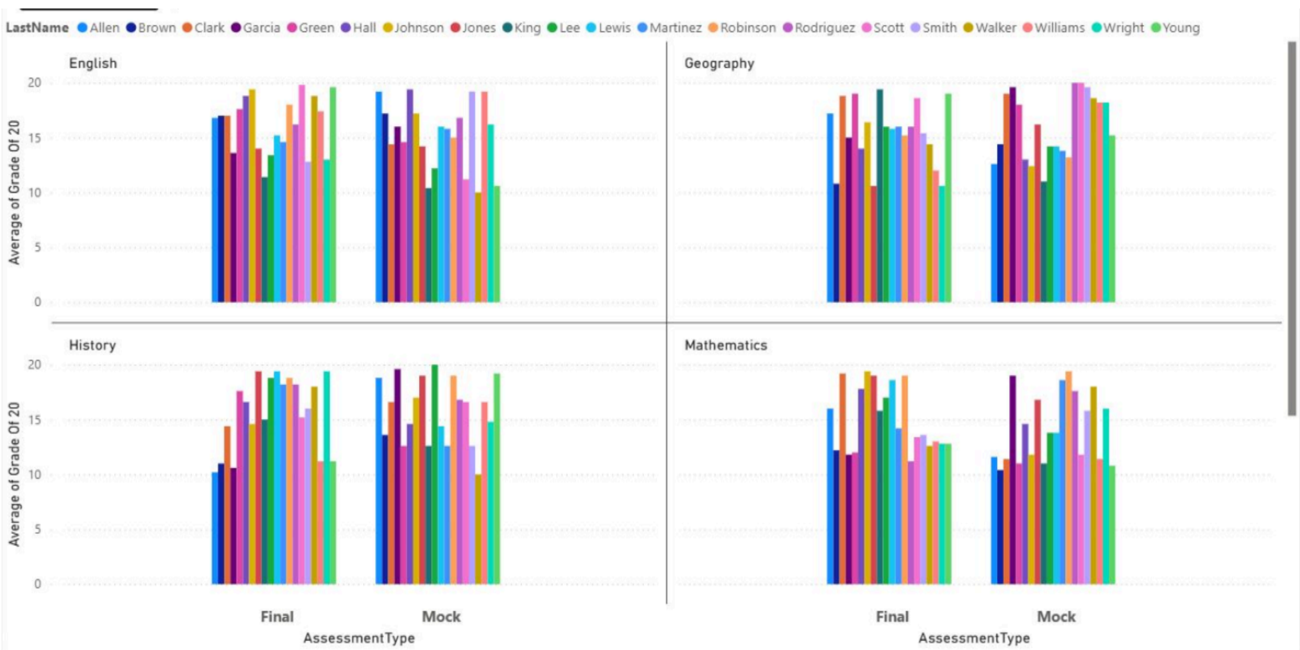


Figure 1 – Comparison of average grades per subject between mock and final exams across all students.

📈 2. Individual Student Performance – Grades by Subject

In this view, we can track the performance of a single student across all subjects.
It shows their grades in detail and helps teachers and parents see exactly where the student is doing well or needs more support.
This is more personal than looking at class averages and allows for focused feedback.

Goal: Monitor each student's progress individually and identify strengths and weaknesses.



📝 *Figure 2 – Detailed grade breakdown for a selected student across all subjects.*

📊 3. Compare a Student's Grades with Class Average (Figure 3)

This visualisation compares an individual student's grades with the class average in each subject.
It helps teachers, parents, and advisors see clearly whether the student is performing above or below the overall class level.
This is useful for understanding the student's relative position in the class and making better decisions about their learning path.

The chart also includes a Decomposition Tree, which presents the same analysis in a different visual way.
Using this tool, we can clearly identify and visualise high-achieving and lower-performing students for each subject, across both mock and final exams.

Goal: Evaluate each student's performance relative to the class average and identify strengths and weaknesses visually.

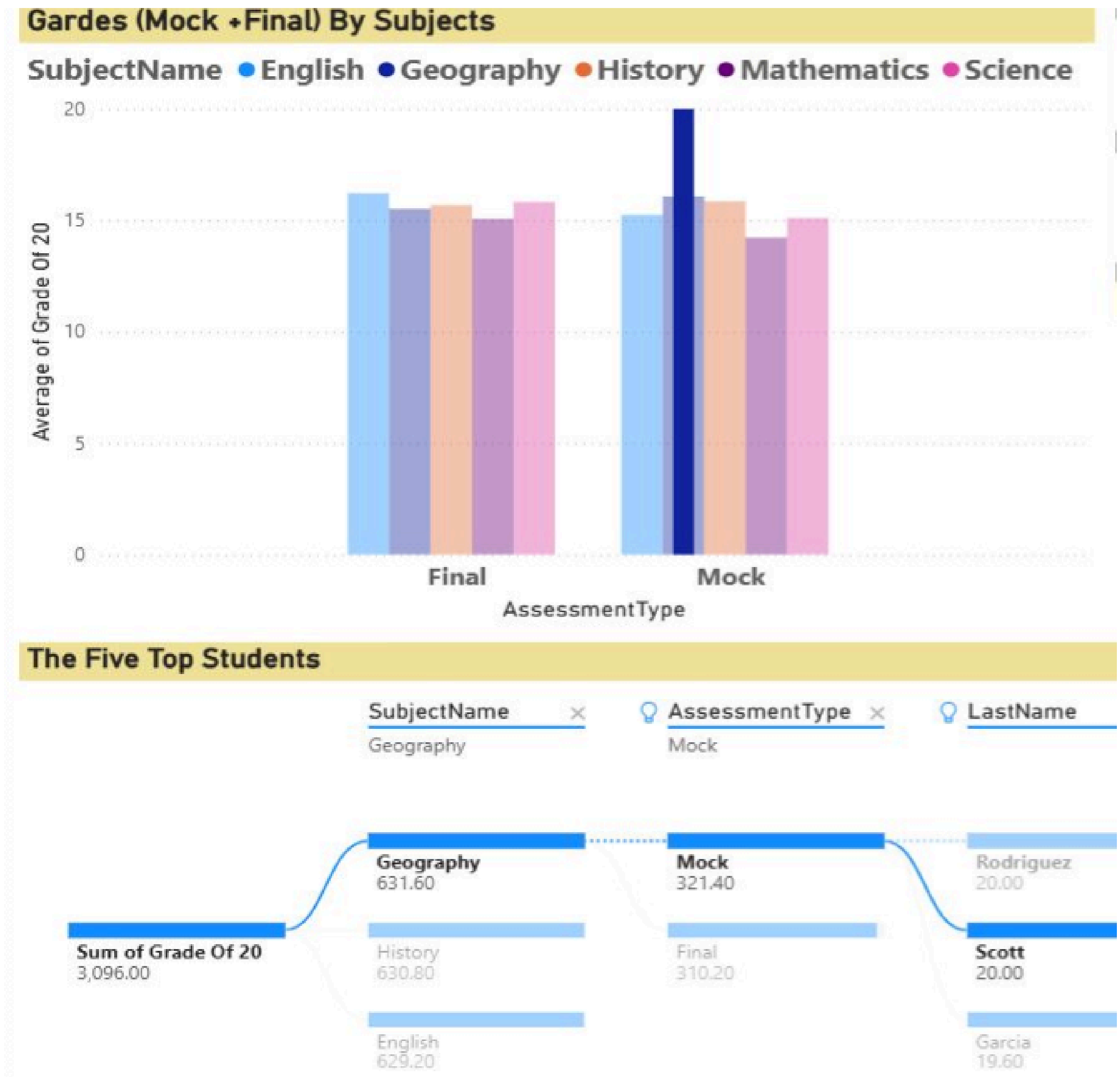


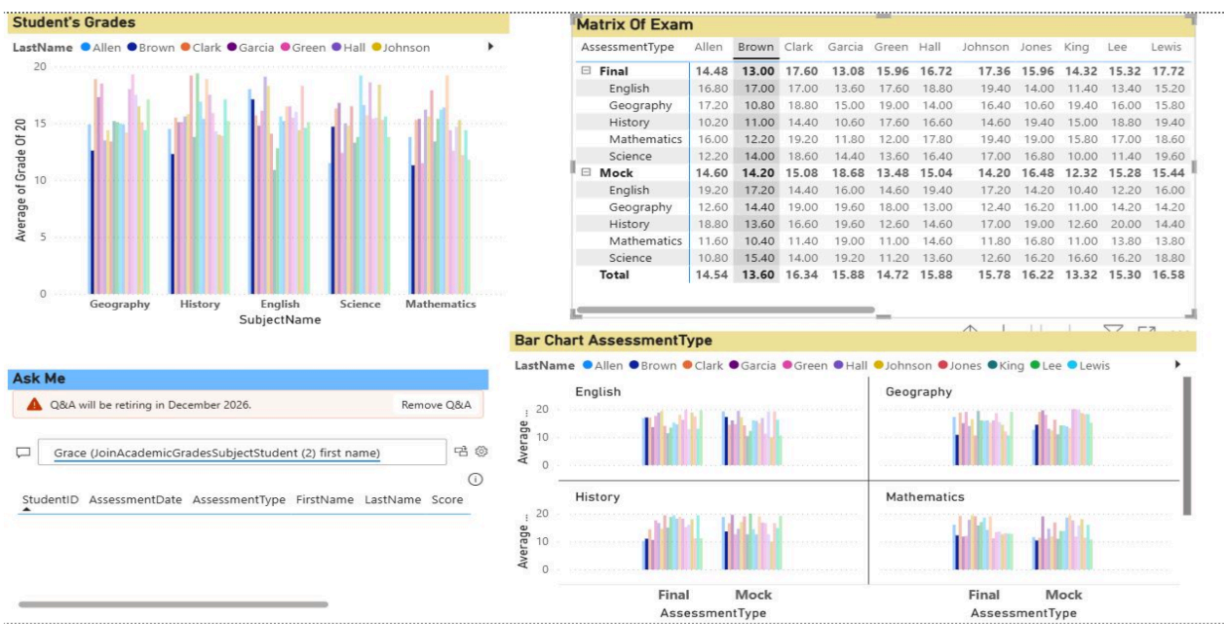
Figure 3 – Comparison of a student's grades with class average, including a Tree Map visualisation.

📊 4. Trace Student's Situation in Different Charts Simultaneously

This visualisation allows us to track a single student's performance across multiple charts at the same time.
Instead of looking at just one chart, we can see the student's data in different views simultaneously, such as grade comparisons, subject breakdowns, and performance trends.
This gives teachers and advisors a more complete and real-time understanding of how the student is doing from different angles.

It also helps to compare the student's performance with their peers while looking at various aspects of their academic record at once.
This multi-chart approach is very useful during parent-teacher meetings or academic counselling sessions because it provides a full picture without switching between different pages or reports.

Goal: Monitor a student's performance from multiple perspectives simultaneously and enable better comparison with peers.



📝 *Figure 4 – Simultaneous tracking of a student's performance across multiple visualisations.*

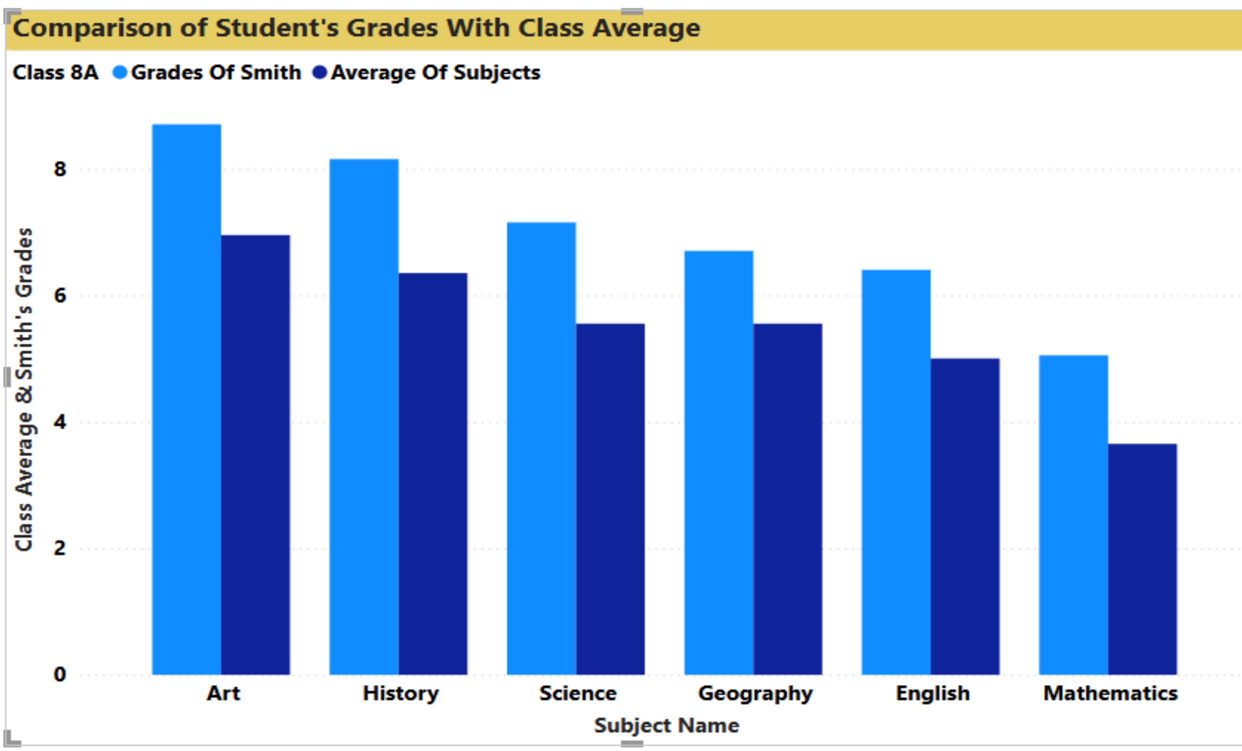
📊 5. Simultaneous Monitoring of Student Performance with Multiple Visualisations and Peer Comparison

This visualisation allows us to track a student's performance, while also comparing them with their classmates.

For example, this student (Smith) is doing very well in Art and History, but their grades in Mathematics and English are lower than the class average. This shows that Smith needs more support in those subjects.

With just one look, teachers, counsellors, and school administrators can immediately understand a student's strengths and weaknesses.
This tool is also very useful for parent-teacher meetings because it clearly shows the student's position compared to the rest of the class, without needing to check multiple reports.

Goal: Monitor student performance from multiple views and compare with peers for better decision-making.



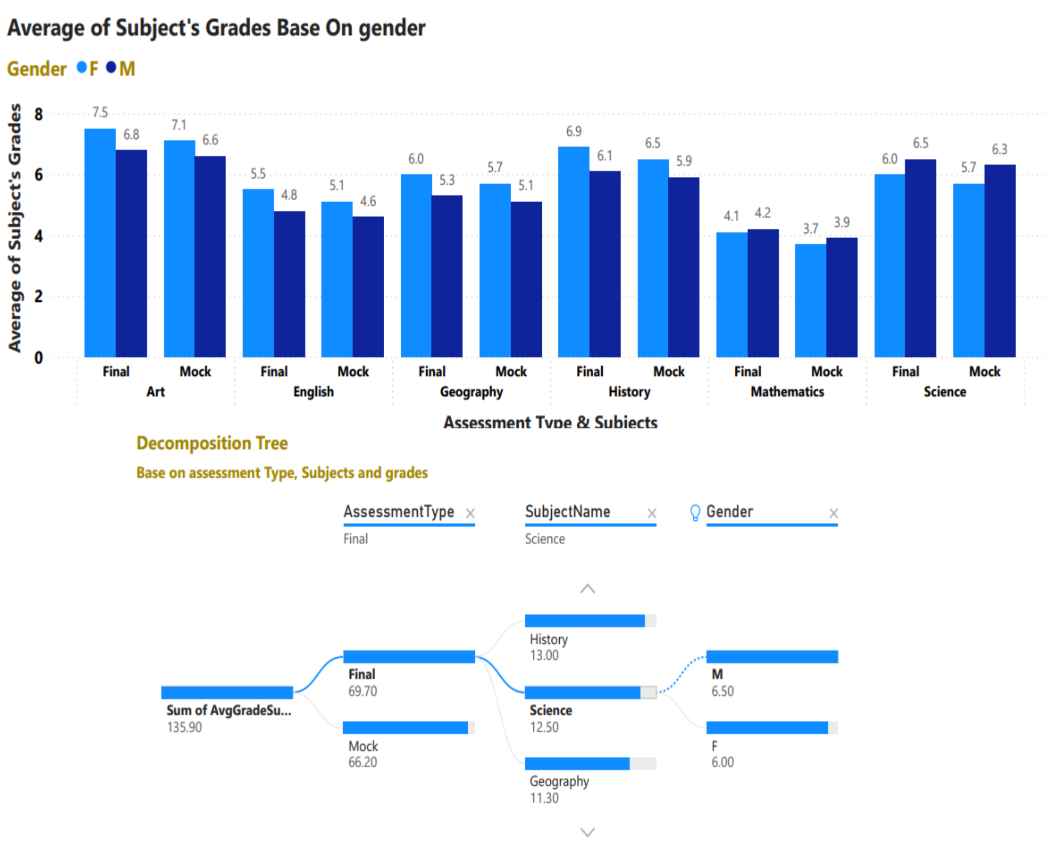
📝 *Figure 5 – Simultaneous tracking of a student's grades compared to class average*

📊 6. Average of Subject's Grades Based on Gender

This visualisation compares the average grades of male and female students in each subject. It helps teachers and school administrators see if there are any significant differences in performance between boys and girls across different subjects.

For example, the chart might show that girls perform better in languages while boys perform better in mathematics.
This information can be used to encourage healthy academic competition between genders and also to adjust teaching methods if needed.

Goal: Compare academic performance between male and female students to identify patterns and promote healthy competition.



📝 Figure 6 Comparison of subject averages by gender.

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📊 7. Combining Discipline with Academic Performance

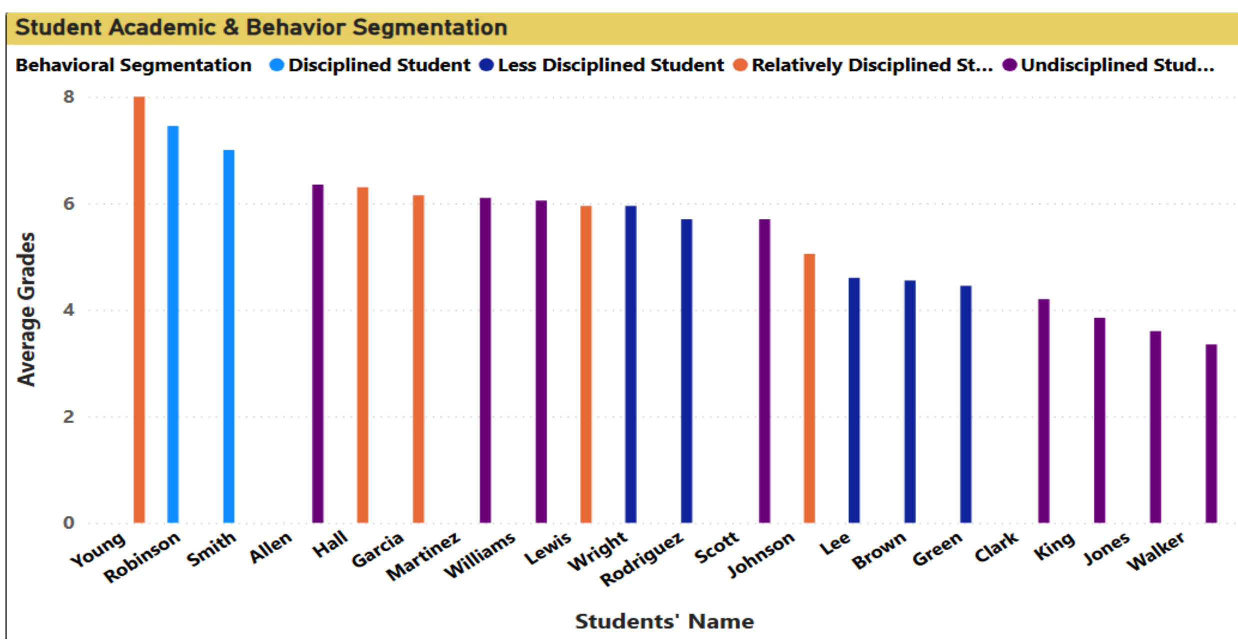
This visualisation combines students' academic grades with their behavioural records.
It helps the school analyse the relationship between discipline and academic success.

In our school, for example, we have two students who are very intelligent and get good grades, but they often behave badly in class. Their disruptive behaviour affects the learning environment and disturbs other students.

The school needs to take action to help these students improve their behaviour.
If we can support them in becoming more disciplined, they will not only stop disrupting the class but also achieve even better academic results in the future.

This type of analysis is valuable because it reminds us that good behaviour and academic success go hand in hand.

Goal: Analyse the connection between student discipline and academic performance, and support students who need behavioural improvement.



📝 *Figure 7 – Integration of discipline records with student grades for behavioural and academic analysis.*